# DENIS SHTEPA

## Research-Oriented System Designer & Team Lead — AI-Native Architecture (KG / MCP / RAG)

Tomsk, Russia | Email: ischademadda@gmail.com | Telegram: @ischademadda | GitHub: github.com/ischademadda | BSc Software Engineering (TPU, 2027, GPA: 3.93/5.0 — Top 10%, ~3.5/4.0)

**Research-oriented System Designer and Team Lead** with 2+ years leading an engineering R&D team (4–7 engineers). Shipped 2 production platforms from 0 to 1: **archmap** (adopted across 4 products / 25 microservice repos at T-Bank, reducing C4 architecture delivery from 3 days to 15 mins) and **OmniSearch Engine** (on-premise multimodal RAG video search deployed at Gazprom Vostok Neft). Focused on AI-native architecture — LLM pipelines with verifiable Knowledge Graphs, MCP-based agent context — and applied systems research. Seeking a Master's in Computer Science / Software Engineering (Fall 2027).

## LEADERSHIP & PROFESSIONAL EXPERIENCE

**Team v3.9 — Applied R&D Lab** **Tomsk, Russia**

*Founder, Team Lead & System Designer (Team: 4–7 engineers)* *Sep 2024 — Present*

### 1. Archon — AI-Native Source of Truth & Enterprise Architecture Platform `[Kotlin, Python, Temporal, PostgreSQL + AGE, Qdrant, LikeC4]`

- **Core Product Vision:** Solved architectural drift by designing an automated Knowledge Graph platform that synchronizes C4 architecture with code and feeds verified context to AI engineering agents via MCP.
- **System Design:** Decomposed 17 logical modules into 3 polyglot services orchestrated by Temporal Polyglot Workflows; authored 7 ADRs closing 100% of technical blockers prior to implementation.
- **Code-to-Architecture Pipeline:** Built a hybrid extraction pipeline: deterministic AST parsing (Tree-sitter) + 2-phase LLM extraction with a strict "Proof-of-Truth" rule (discarding unproven graph edges), cutting LLM tokens by 90%.
- **Scalable Graph Storage:** Benchmarked multi-tenant Graph-per-Tenant isolation on PostgreSQL 16 + Apache AGE across 50 schemas: Impact Analysis traversal <15ms at depth 4 (NFR <100ms).

### 2. OmniSearch Engine — On-Premise Multimodal Video RAG System `[Kotlin Ktor, Go, Python Whisper/CLIP, Qdrant, MinIO S3]`

- **System Architecture:** Separated Control Plane (Kotlin Ktor) and Compute Plane (Go worker pool, Python ML nodes); defined OpenAPI 3.0 contracts and asynchronous media processing finite-state machine.
- **Industrial Adoption:** Delivered on-premise semantic video search (Whisper audio transcription + CLIP frame embeddings); deployed at **Gazprom Vostok Neft** for safety compliance monitoring.

**T-Bank** **Moscow, Russia (Remote)**

*System Analyst (Junior+, Full-Cycle System Design)* *Apr 2026 — Aug 2026 (Internship, Full-Cycle System Design)*

- **archmap Platform (Founder & PO):** Initiated and launched an automated C4 architecture generator from scratch. Combined AST extraction, LLMs, and LikeC4 with CI/CD integration. Reduced documentation delivery from 1–3 days to 15 mins; approved by management and scaled across 4 products (~25 repos).
- **Cross-System Integrations:** Designed end-to-end specifications for co-branding features across 3 external systems with backward-compatible REST APIs; eliminated delivery blockers at the specification phase.
- **Operational Efficiency:** Refactored on-call runbooks across 145 incidents, reducing mean response time by 40%.

## CORE COMPETENCIES & SKILLS

**Product & Architecture:** System Design, C4 Model, LikeC4, OpenAPI/REST, ADR/RFC, BPMN, Use Cases, NFR, Impact Analysis.

**Distributed Systems & AI:** Microservices, Event-Driven Architecture, Temporal, Knowledge Graphs (Apache AGE), Vector DBs (Qdrant), RAG, Model Context Protocol (MCP).

**Languages & Technologies:** Kotlin, Go, Python, TypeScript, PostgreSQL, Redis, Docker, Git | English C1 (IELTS 7.5 expected Dec 2026), Chinese HSK (learning).